Ordinary and spectral, compared

Both write A as a diagonal matrix conjugated by an eigenvector matrix. Symmetry upgrades every line of the comparison — the eigenvectors, the eigenvalues, the guarantee, and the cost of inverting.

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KINDS

TYPE	CONDITION	DIAGONALIZING MATRIX	EIGENVECTOR COLUMNS	ENTRIES OF D	WHEN IT EXISTS
ANY DIAGONALIZABLE MATRIX					
 **Ordinary**	$A = PDP^{-1}$	P invertible — nothing more	linearly independent	real or complex	iff $m_g(\lambda) = m_a(\lambda)$ for every λ
REAL SYMMETRIC MATRICES					
 **Spectral**	$A = QDQ^{\mathsf{T}}$	Q orthogonal — $Q^{-1} = Q^{\mathsf{T}}$	orthonormal	all real	always, for every real symmetric A

Every improvement traces to one hypothesis. Symmetry forces real eigenvalues, forces eigenvectors for distinct eigenvalues to be orthogonal, and forces enough of them to exist — so the spectral theorem is not a special case so much as the case where nothing can go wrong.